

CSA0921 - Programming in Java

CO3 - AT1 - Debugging and Optimization

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1.

The screenshot shows the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" with "JAVA PROGRAMMING" below it. The user is logged in as "Vidhushini T S" with ID "192511061IRC". The current section is "CO3- AT1 - DEBUGGING AND OPTIMIZATION". On the left, there is a "QUESTIONS" sidebar with five questions listed: Q1 (Debug Nested Exception Handling, 10 marks), Q2 (Debug Exception Propagation, 10 marks), Q3 (Debug Multiple Runtime Exceptions, 10 marks), Q4 (Debug Finally Execution, 10 marks), and Q5 (Debug the NullPointerException, 10 marks). Q1 is selected. The main area displays the question "Q1 Debug Nested Exception Handling" with a "10 marks" label. The code provided is as follows:

```
import java.util.Scanner;

public class Debug1 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int arr[] = {20,40,60,80};

        int index = sc.nextInt();
        int divisor = sc.nextInt();

        try {
            int value = arr[index];
            try {
                int result = value/divisor;
                System.out.println(result);
            }
            catch(ArrayIndexOutOfBoundsException e) {
                System.out.println("Invalid Index");
            }
        }
        catch(ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}

Debugging Task
Identify why some exceptions are never caught.
```

The bottom of the screen shows a Windows taskbar with various application icons and system information: 35°C Sunny, Search, and the date/time 14:33 11-08-2026.

Output:

The screenshot shows the SIMATS Online Programming Lab interface for a debugging task. The page title is "SIMATS Online Programming Lab" with "JAVA PROGRAMMING" below it. The user is logged in as "Vidhushini T S" with ID "192511061IRC". The current section is "CO3- AT1 - DEBUGGING AND OPTIMIZATION". The main area displays a "Debugging Task" with the instruction: "Find and fix the bugs in the code shown above. Write your corrected code in the editor below." The code provided is as follows:

```
1 import java.util.Scanner;
2 public class Debug1{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int arr[]={20,40,60,80};
6         int index=sc.nextInt();
7         int divisor=sc.nextInt();
8         try{
9             int value=arr[index];
10            try{
11                int result=value/divisor;
12                System.out.println(result);
13            }
14            catch(ArithmeticException e){
15                System.out.println("Division by zero");
16            }
17        }
18        catch(ArrayIndexOutOfBoundsException e){
19            System.out.println("Invalid index");
20        }
21    }
22 }
```

Below the code editor, there are buttons for "Run", "Save", and "Submit Fix". To the right of the code editor, there is a "Test Input" field with the value "1 0", an "Output" field showing "Division by zero", and a "Test Cases" section with three cases: "Test Case 1", "Test Case 2", and "Test Case 3". The bottom of the screen shows a Windows taskbar with various application icons and system information: 35°C Sunny, Search, and the date/time 14:33 11-08-2026.

2.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user "Vidhushini T S". The main section is titled "CO3- AT1 - DEBUGGING AND OPTIMIZATION". On the left, a "QUESTIONS" sidebar lists five questions, with the first two marked as answered. The main area displays "Q2 Debug Exception Propagation" with a 10-mark rating. It contains a Java code snippet for a calculator and a debugging task: "Find and fix the bugs in the code shown above. Write your corrected code in the editor below." The code is as follows:

```
import java.util.Scanner;

class Calculator{
    static int divide(int a,int b){
        return a/b;
    }

    static void display(int x){
        System.out.println("Answer = "+x);
    }
}

public class Debug2{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int a=sc.nextInt();
        int b=sc.nextInt();
        Calculator.display(Calculator.divide(a,b));
        System.out.println("Execution Completed");
    }
}

Debugging Task
Prevent abnormal termination.
Ensure "Execution Completed" is always printed.
```

Output:

The screenshot shows the SIMATS Online Programming Lab interface with the code editor and test results. The code is as follows:

```
1 import java.util.Scanner;
2 class Calculator{
3     static int divide(int a,int b){
4         return a/b;
5     }
6     static void display(int x){
7         System.out.println("Answer = "+x);
8     }
9     public static void main(String[] args){
10        Scanner sc=new Scanner(System.in);
11        int a=sc.nextInt();
12        int b=sc.nextInt();
13        try{
14            Calculator.display(Calculator.divide(a,b));
15        }catch(ArithmeticException e){
16            System.out.println("Cannot divide by zero");
17        }
18        System.out.println("Execution Completed");
19    }
20 }
```

The test input is "20 5". The output is "Answer = 4" and "Execution Completed". The test cases are listed as "Test Case 1", "Test Case 2", and "Test Case 3".

3.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the page title "SIMATS Online Programming Lab", and a user profile for "Vidhushini T S". The main content area is titled "CO3- AT1 - DEBUGGING AND OPTIMIZATION". On the left, a "QUESTIONS" sidebar lists five items, with the third item, "Q3 Debug Multiple Runtime Exceptions", selected. The main panel displays the details for this question, which is worth 10 marks. It includes a "Debugging Task" description: "The program may generate two different runtime exceptions. Modify the program so that each exception is handled separately." Below this, a "Your Fixed Code" section shows a Java code snippet with several errors. To the right of the code is a "Test Input" field containing the value "5" and an "Output" field showing "Index Out of bounds". At the bottom of the interface, a taskbar displays the system date and time as 11-08-2026, 14:57, along with weather information (35°C Sunny) and various application icons.

Team 3. DS report - Google Do... SIMATS Online Lab... Untitled document - Google Do... Ask Gemini

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SIMATS ENGINEERING SIMATS Online Programming Lab JAVA PROGRAMMING Vidhushini T S 192510619C

CO3- AT1 - DEBUGGING AND OPTIMIZATION

QUESTIONS

- Q1 Debug Nested Exception Handling 10 marks
- Q2 Debug Exception Propagation 10 marks
- Q3 Debug Multiple Runtime Exceptions 10 marks
- Q4 Debug Finally Execution 10 marks
- Q5 Debug the NullPointerException 10 marks

3/5 answered

Q3 Debug Multiple Runtime Exceptions 10 marks

import java.util.Scanner;

public class Debug3 {

public static void main(String args[]) {

Scanner sc=new Scanner(System.in);

int arr[]={12,24,36,48};

String number=sc.next();

int index=Integer.parseInt(number);

System.out.println(arr[index]);

}

}

Debugging Task

The program may generate two different runtime exceptions.

Modify the program so that each exception is handled separately.

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug3{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int arr[]={12,24,36,48};
6         try{
7             String number=sc.next();
8             int index=Integer.parseInt(number);
9             System.out.println(arr[index]);
10        } catch (NumberFormatException e){
11            System.out.println("Invalid number");
12        } catch (ArrayIndexOutOfBoundsException e){
13            System.out.println("Index Out of bounds");
14        }
15    }
16 }
```

Test Input

5

Output

Index Out of bounds

35°C Sunny 11-08-2026 14:57

Output:

This screenshot shows the same SIMATS Online Programming Lab interface after the code has been corrected. The "Your Fixed Code" section now contains a Java program that uses try-catch blocks to handle both "NumberFormatException" and "ArrayIndexOutOfBoundsException". The "Test Input" field still contains "5", and the "Output" field now displays "Index Out of bounds". Below the code editor, there are buttons for "Run", "Save", and "Submit Fix". To the right of the code, a "Test Cases" section shows three cases, each with a status indicator. The bottom of the interface remains the same, showing the system date and time as 11-08-2026, 14:57, and weather information (35°C Sunny).

Team 3. DS report - Google Do... SIMATS Online Lab... Untitled document - Google Do... Ask Gemini

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SIMATS ENGINEERING SIMATS Online Programming Lab JAVA PROGRAMMING Vidhushini T S 192510619C

CO3- AT1 - DEBUGGING AND OPTIMIZATION

QUESTIONS

- Q1 Debug Nested Exception Handling 10 marks
- Q2 Debug Exception Propagation 10 marks
- Q3 Debug Multiple Runtime Exceptions 10 marks
- Q4 Debug Finally Execution 10 marks
- Q5 Debug the NullPointerException 10 marks

3/5 answered

Q5 Debug the NullPointerException 10 marks

Debugging Task

The program may generate two different runtime exceptions.

Modify the program so that each exception is handled separately.

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 public class Debug3{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int arr[]={12,24,36,48};
6         try{
7             String number=sc.next();
8             int index=Integer.parseInt(number);
9             System.out.println(arr[index]);
10        } catch (NumberFormatException e){
11            System.out.println("Invalid number");
12        } catch (ArrayIndexOutOfBoundsException e){
13            System.out.println("Index Out of bounds");
14        }
15    }
16 }
```

Test Input

5

Output

Index Out of bounds

Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

Run Save Submit Fix

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35°C Sunny 11-08-2026 14:57

4.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user "Vidhushini T S". The main section is titled "CO3- AT1 - DEBUGGING AND OPTIMIZATION". On the left, a "QUESTIONS" sidebar lists five questions, with the first four marked as answered. The main area displays "Q4 Debug Finally Execution" with a code editor containing the following Java code:

```
public class Debug4 {  
    static int calculate(int a,int b){  
  
    try{  
        return a/b;  
    }  
    finally{  
        System.out.println("Resources Released");  
    }  
}  
  
public static void main(String args[]){  
    System.out.println(calculate(20,0));  
}  
}
```

Below the code, the task instructions are: "Debugging Task. Prevent program termination. Ensure the finally block executes while handling the exception correctly."

Output:

The screenshot shows the SIMATS Online Programming Lab interface with the "Debugging Task" instructions: "Find and fix the bugs in the code shown above. Write your corrected code in the editor below." The "Your Fixed Code" section displays the corrected Java code:

```
1 public class Debug4{  
2     static int calculate(int a, int b){  
3         try{  
4             return a/b;  
5         }catch(ArithmeticException e){  
6             System.out.println("Division by zero");  
7             return 0;  
8         }finally{  
9             System.out.println("Resources Released");  
10        }  
11    }  
12    public static void main(String[] args){  
13        System.out.println(calculate(20,5));  
14    }  
15 }
```

The "Test Input" section shows "stdin input (optional)". The "Output" section displays the result: "Resources Released" followed by "4". The "Test Cases" section lists "Test Case 1", "Test Case 2", and "Test Case 3". At the bottom, there are buttons for "Run", "Save", and "Submit Fix".

5.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the subject "JAVA PROGRAMMING". The user is logged in as "Vidhushini T S" with ID "192511061RC". The main section is titled "CO3- AT1 - DEBUGGING AND OPTIMIZATION". On the left, there is a list of questions: Q1 (Debug Nested Exception Handling, 10 marks), Q2 (Debug Exception Propagation, 10 marks), Q3 (Debug Multiple Runtime Exceptions, 10 marks), Q4 (Debug Finally Execution, 10 marks), and Q5 (Debug the NullPointerException, 10 marks). The selected question is Q5. The task description is "Debug the NullPointerException" with 10 marks. The code provided is:

```
public class Debug4 {  
    public static void main(String[] args) {  
        String name = null;  
        System.out.println(name.length());  
    }  
}
```

The expected task is to prevent the program from crashing. A debugging task instruction says: "Find and fix the bugs in the code shown above. Write your corrected code in the editor below." The "Your Fixed Code" section shows the corrected code:

```
1 public class Debug4 {  
2     public static void main(String[] args) {  
3         String name="Java";  
4         if (name!=null) {  
5             System.out.println("Length =" + name.length());  
6         } else {  
7             System.out.println("String is Null");  
8         }  
9     }  
10 }
```

The "Test Input" field is empty, and the "Output" field is also empty.

Output:

The screenshot shows the same SIMATS Online Programming Lab interface, but now the "Output" field displays the result of the code execution: "Length =4". The "Test Cases" section shows three test cases, all of which are marked as passed. The "Run" button is highlighted in green, indicating that the code has been successfully executed. The "Submit Fix" button is also visible.